Finding the Next Best Step

Students know what they want to become, not what to do next. A few smart questions → one concrete next step, or a named mentor when the app is unsure.

INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY
HYDERABAD

Healthcare & Social Impact Student Project Showcase

THE PROBLEM & CONTEXT

"I want to become a data analyst but I don't know where to start."
— our example student

9%	of rural homes have a computer or laptop	**<15%**	finish non-live online courses
70%	of low-income students disengage from recorded courses	**16.6%**	of colleges cross 75% placement

- **Unknown unknowns:** can't search for what you lack
- **Long forms** on shared, slow phones → drop-off

Statistics: The Barabari Collective, barabaricollective.org

OUR APPROACH

Sprint 0 · Understand
Mapped the problem with Barabari

Sprint 1 · Design
11 Figma frames + a 4-part engine

Sprint 2 · Build
Android app, ~7,000 lines of Kotlin

Demo · Listen
5 refinements from Barabari

We assumed: *one step beats a full list, and the blocker is information, not fees.*

Why existing options stop short

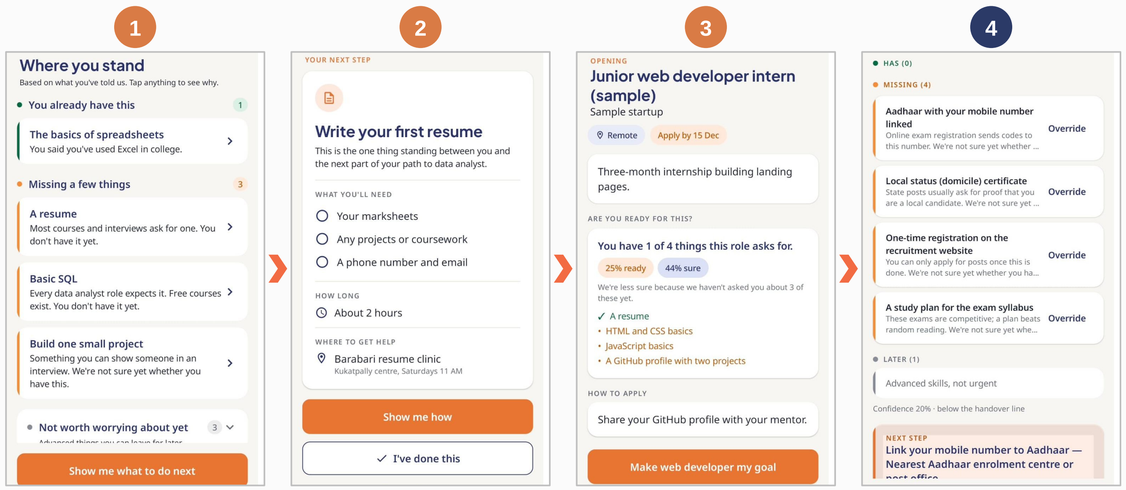
- **Govt portals** — jargon, long forms
- **Chatbots** — confident & wrong
- **YouTube** — generic, not you
- **NGO casework** — not there at 9 pm

Our design choices

- Design for the student
- One question at a time
- Rules decide gaps
- Hand over, don't guess

OUR SOLUTION

An Android app with two sides: students find their next step, caseworkers stay in the loop.



See where you stand · **Get ONE next step** · **Job match: how ready** · **Caseworker overrides**

HOW IT WORKS

S1 · Dual parser — AI
resumes & job listings → one schema

S2 · Question picker — Rules + AI wording
asks the most useful question

S3 · Matcher — Rules
top roles + shortest path for each

S4 · Confidence gate — Rules
sure enough to answer?

- ≥ 70% sure → **One next step**
- Unsure → **Goal-matched mentor**

Kotlin · Compose · Room · OpenAI / Groq

WHAT WE LEARNED

17 engine unit tests

Offline Support — If AI is down

Barabari feedback we acted on

- Openings show how ready you are
- Two-way match: resumes ↔ jobs
- Mentor matched to the kind of work
- A reason behind every item

Limitations

- Tested only with demo personas
- Without AI, questions get simpler

Design for Social Innovation

IMPACT AND WHATS NEXT

- **One action, not a catalogue**
- **A gap becomes a task, not a rejection**
- **Mentors focus on students, not spreadsheets**

What's next

1. Barabari reviews the roles
2. Shared backend, not one phone
3. API key moves to a server
4. Separate documents from skills
5. Pilot with Barabari students

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REFERENCES

The Barabari Collective (barabaricollective.org)
myScheme (myscheme.gov.in)
OGD Platform India (data.gov.in)
Mortati & Villari, Design and Social Innovation, ServDes 2014

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